- Loading data: Drive, Sheets, and Google Cloud Storage
- Charts: visualizing data
- Getting started with BigQuery

Machine Learning

These are a few of the notebooks related to Machine Learning, including Google's online Machine Learning course. See the full course website for more.

- Intro to Pandas DataFrame
- Intro to RAPIDS cuDF to accelerate pandas
- Getting Started with cuML's accelerator mode

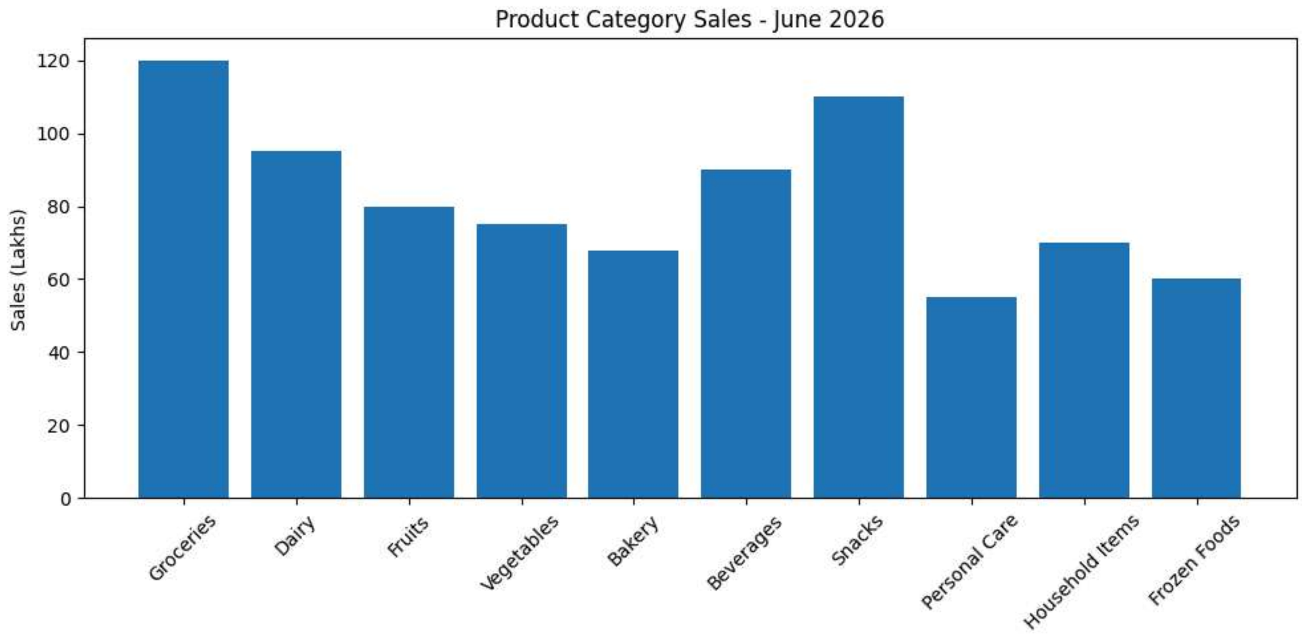
Using Accelerated Hardware

- Train a CNN to classify handwritten digits on the MNIST dataset using Flax NNX API
- Train a Vision Transformer (ViT) for image classification with JAX
- Text classification with a transformer language model using JAX

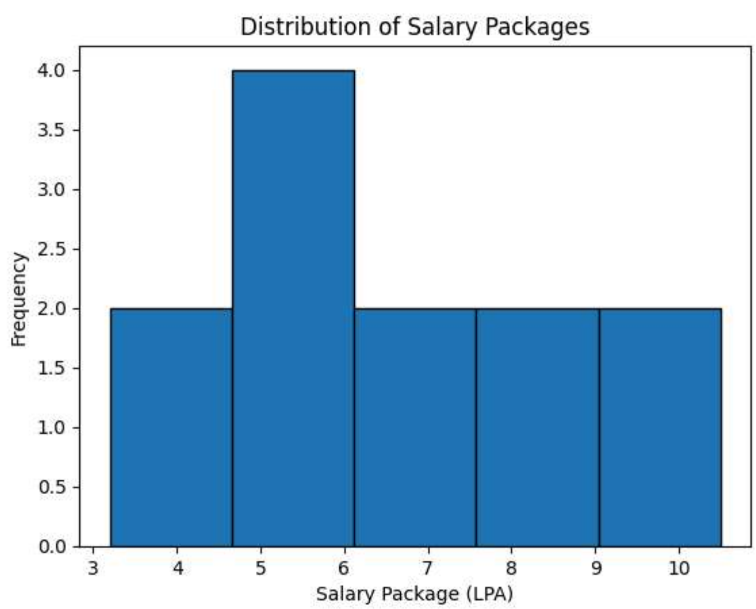
Featured examples

- Train a miniGPT language model with JAX AI Stack
- LoRA/QLoRA finetuning for LLM using Tunix
- Parameter-efficient fine-tuning of Gemma with LoRA and QLoRA
- Loading Hugging Face Transformers Checkpoints
- 8-bit Integer Quantization in Keras
- Float8 training and inference with a simple Transformer model
- Pretraining a Transformer from scratch with KerasHub
- Simple MNIST convnet
- Image classification from scratch using Keras 3
- Image Classification with KerasHub

```
import matplotlib.pyplot as plt

categories = ["Groceries","Dairy","Fruits","Vegetables","Bakery",
              "Beverages","Snacks","Personal Care",
              "Household Items","Frozen Foods"]
sales = [120,95,80,75,68,90,110,55,70,60]

plt.figure(figsize=(10,5))
plt.bar(categories, sales)
plt.xticks(rotation=45)
plt.ylabel("Sales (Lakhs)")
plt.title("Product Category Sales - June 2026")
plt.tight_layout()
plt.show()
```

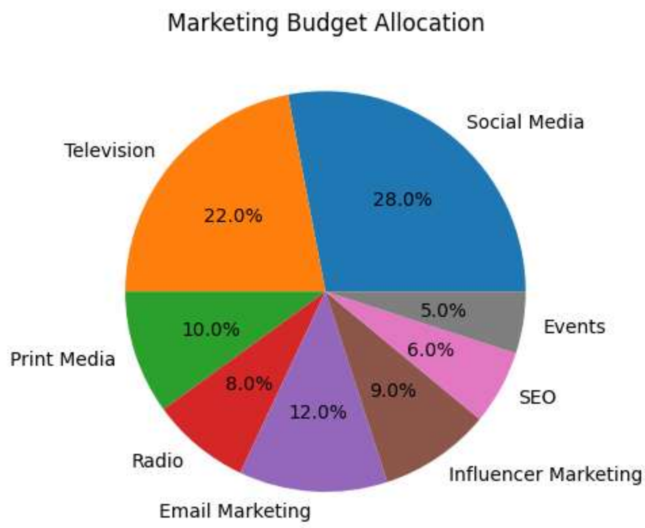



```
import matplotlib.pyplot as plt

salary = [3.2,4.5,5.0,6.8,7.2,8.5,4.8,5.5,6.0,7.8,9.2,10.5]

plt.hist(salary, bins=5, edgecolor='black')
plt.xlabel("Salary Package (LPA)")
plt.ylabel("Frequency")
plt.title("Distribution of Salary Packages")
plt.show()
```

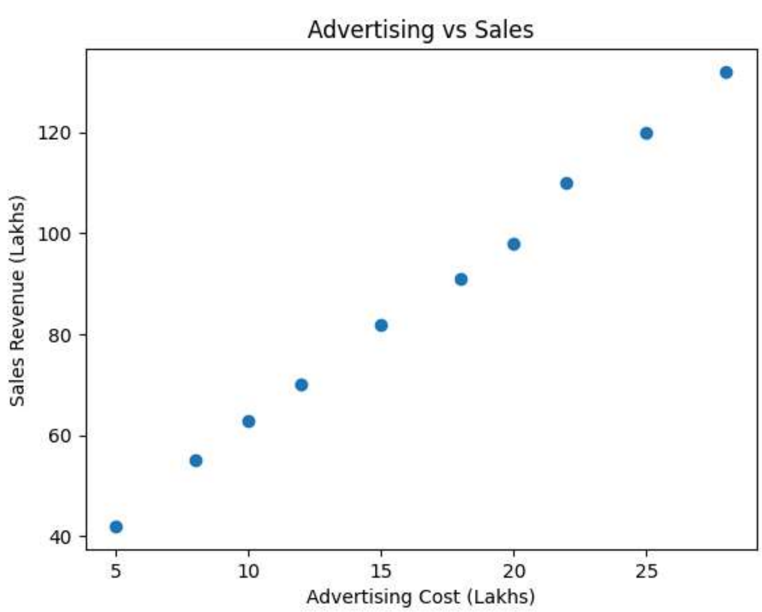



```
import matplotlib.pyplot as plt

channels = ["Social Media","Television","Print Media","Radio",
            "Email Marketing","Influencer Marketing","SEO","Events"]
budget = [28,22,10,8,12,9,6,5]

plt.pie(budget, labels=channels, autopct='%1.1f%%')
```

```
plt.title("Marketing Budget Allocation")
plt.show()
```




```
import matplotlib.pyplot as plt

ad_cost = [5,8,10,12,15,18,20,22,25,28]
sales = [42,55,63,70,82,91,98,110,120,132]

plt.scatter(ad_cost, sales)
plt.xlabel("Advertising Cost (Lakhs)")
plt.ylabel("Sales Revenue (Lakhs)")
plt.title("Advertising vs Sales")
plt.show()
```




```
import matplotlib.pyplot as plt

months = ["Jan","Feb","Mar","Apr","May","Jun",
          "Jul","Aug","Sep","Oct","Nov","Dec"]
energy = [420,450,490,530,560,600,580,590,610,630,590,540]

plt.plot(months, energy, marker='o')
plt.xlabel("Month")
plt.ylabel("Energy Generated (MWh)")
```

```
plt.title('Monthly Solar Energy Generation - 2025')
plt.grid(True)
plt.show()
```